Advanced Topology

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Everything comes in time to him who knows how to wait.

—Leo Tolstoy, [Tol69]

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1.1.1 Inadequacy of sequences

In a metric space (X, d) there always exists a convergent sequence. This is easy to see. Let x be any point in the metric space X, then for every $r > 0$ there is an open ball $B(x, r)$ centered at x of radius r. Put $r = 1/n$, then we have a sequence of balls $\{B(x, 1/n)\}_n$ centered at $x \in X$. Now we just choose $x_n \in B(x, 1/n)$, then we have a sequence $\{x_n\}_n$ and certainly $x_n \to x$: choose $M \in \mathbb{N}$ s.t. $1/M < r$, then

$$d(x_n, x) < \frac{1}{n} \leqslant \frac{1}{M} < r \quad \forall n \geqslant M.$$

So in a metric space, convergent sequences always exist and determine the *topology* on it. The point a sequence converges to is its *limit point*, and the set of points in $S \subset X$ with its limit points yields its *closure* $\overline{S}$. If $S = \overline{S}$, then S is a *closed set* and $X \setminus S$ is an *open set*. Further, a function $f : X \to X$ is continuous at $x \in X$ if for every convergent sequence $x_n \to x$ we have $f(x_n) \to f(x)$. So convergent sequences in X characterise the *topology* on X - the open sets, the closed sets, as well as the continuity of functions on X.

However, when dealing with more general topological spaces, the above argument - in particular the part where we took a sequence of balls centered at x, only works if X has a countable local base, i.e., X has to be *first-countable*.

Let $(X, \mathcal{T})$ be a topological space. A *sequence* is just a function $f : \mathbb{N} \to X$. Usually, the point $f(n)$ is denoted x_n, and the sequence is written $\{x_n\}_n$. We say $x_n \to x$ if for any open set U containing x, there exists an $M \in \mathbb{N}$ s.t. $x_n \in U$ for all $n \geqslant M$. Now, w.r.t. $\leqslant$, $\mathbb{N}$ has the following properties,

($\preceq$1) $a \leqslant a$ (reflexivity)

($\preceq$2) $a \leqslant b$ and $b \leqslant c \implies a \leqslant c$ (transitivity)

($\preceq$3) $a \leqslant b$ and $b \leqslant a \implies a = b$ (antisymmetry)

($\preceq$4) $a \leqslant b$ or $b \leqslant a$ (strongly connected/totality)

The first two properties define a *preorder*, the third defines a *partially ordered set* or *poset*, it is the fourth which makes it a *totally ordered set* or *chain*.

The complex numbers $\mathbb{C}$ cannot be totally ordered, but it can be partially ordered as follows:

$$a + ib \leqslant c + id, \text{ if } a \leqslant b, c \leqslant d$$

which is the same as the following partial order on $\mathbb{R}^2$,

$$(a, b) \leqslant (c, d), \text{ if } a \leqslant b, c \leqslant d.$$

This is not strongly connected, for example $7 + 2i$ and $3 + 5i$ have no relation between them. But there is an element, such as $7 + 5i$, such that $7 + 2i, 3 + 5i \leqslant 7 + 5i$. In fact, there is always an element that is a common upper bound for any two such incomparable elements in $\mathbb{C}$, $\mathbb{R}^2$ or any other poset.

For any pseudometric space (X, d), we drop the requirement $d(x, y) = 0 \implies x = y$. So a pseudometric space may not be a poset - in particular, antisymmetry need not hold. Also recall,

Theorem. Let $(X, \mathcal{T})$ be a T_2 (*Hausdorff*) space. Then every sequence in X converges to a unique limit.

To make the above result about sequences an *if and only if* statement, we'll have to further assume that X is first-countable.

1.1.2 Nets

To summarise: we want to generalise the notion of a sequence for use in more general topological spaces, retaining the idea of ordering a collection of points of X by mapping some ordered set into X while significantly relaxing the conditions on the ordered sets we will allow - in particular, the totality of the order can be dispensed with provided we supply some other way of giving a definite "positive orientation" to our ordered sets. This leads naturally to the notion of a *directed set*.

Definition 1.1 (Directed set). A set $(\Lambda, \leqslant)$ is a **directed set** if

(Λ1) $\lambda \leqslant \lambda$ for all $\lambda \in \Lambda$ (*reflexive*)

(Λ2) if $\lambda_1 \leqslant \lambda_2$ and $\lambda_2 \leqslant \lambda_3$ then $\lambda_1 \leqslant \lambda_3$, (*transitive*)

(Λ3) if $\lambda_1, \lambda_2 \in \Lambda$ then there is some $\lambda_3 \in \Lambda$ s.t. $\lambda_1, \lambda_2 \leqslant \lambda_3$. (*common upper bound*)

Example 1.2. Every chain (e.g., $\mathbb{N}, \mathbb{Q}, \mathbb{R}$ with the usual order) is clearly a directed set.

Example 1.3. Every poset is a directed set, e.g. $\mathbb{C}$ or $\mathbb{R}^2$ with the partial order defined in section 1.1.1.

Example 1.4. For any set $X \neq \varnothing$, the power set $\mathcal{P}(X)$ is a directed set with respect to the partial order $\subseteq$ which can be defined as either,

$$A \leqslant B \text{ if } A \subset B$$

or

$$A \leqslant B \text{ if } A \supset B.$$

In the first case, the common upper bound of A and B is $A \cup B$, while in the second case it is $A \cap B$.

Example 1.5. Every topology $\mathcal{T}$ itself is a directed set with respect to the partial order $\subset$, or more usefully $\supset$.

Example 1.6. If X is a topological space and $x \in X$, then the *open nbhood*[a] *basis*

$$\mathcal{B}_x = \{U \in \mathcal{T} : x \in U\}$$

is a directed set with respect to the partial order $\subset$, or more usefully $\supset$. More generally, the *nbhood/local basis* of x,

$$\mathcal{N}_x = \{G \subset X : G \supset U \text{ for some } U \in \mathcal{B}_x\}$$

is a directed set w.r.t. $\subset$ where $U \leqslant V$ if $U \supset V$, as we always want our nbhoods to be as small as possible.

[a] We'll abbreviate neighbourhood as nbhood.

Example 1.7. If $(\Lambda_1, \leqslant_1)$ and $(\Lambda_2, \leqslant_2)$ are directed sets, then $(\Lambda_1 \times \Lambda_2, \leqslant)$ is a directed set, where

$$(a, b) \leqslant (x, y) \text{ if } a \leqslant_1 x \text{ and } b \leqslant_2 y.$$

Now we can generalise the notion of a sequence by replacing the positive integers $\mathbb{N}$ with a directed set Λ.

Definition 1.8 (Net). A **net** is a function $s : \Lambda \to X$, where Λ is a directed set. We denote the point $s(\lambda) = s_\lambda$ and the net by $\{s_\lambda\}_{\lambda \in \Lambda}$ or simply $\{s_\lambda\}_\lambda$.

For a sequence $\{x_n\}_n$ converging to x, all but finitely many points of the sequence are contained in any open set U containing x. So for a sequence, we have all but finitely many indices $n \geqslant n_0$. But for a net, the directed set may have infinitely many elements that are not comparable to λ_0, even if there are infinitely many elements $\lambda \geqslant \lambda_0$.

Geometrically, we can visualise the positive integers $\mathbb{N}$ as forming a line, whilst the directed set $\mathbb{N} \times \mathbb{N}$ forms a branching tree.

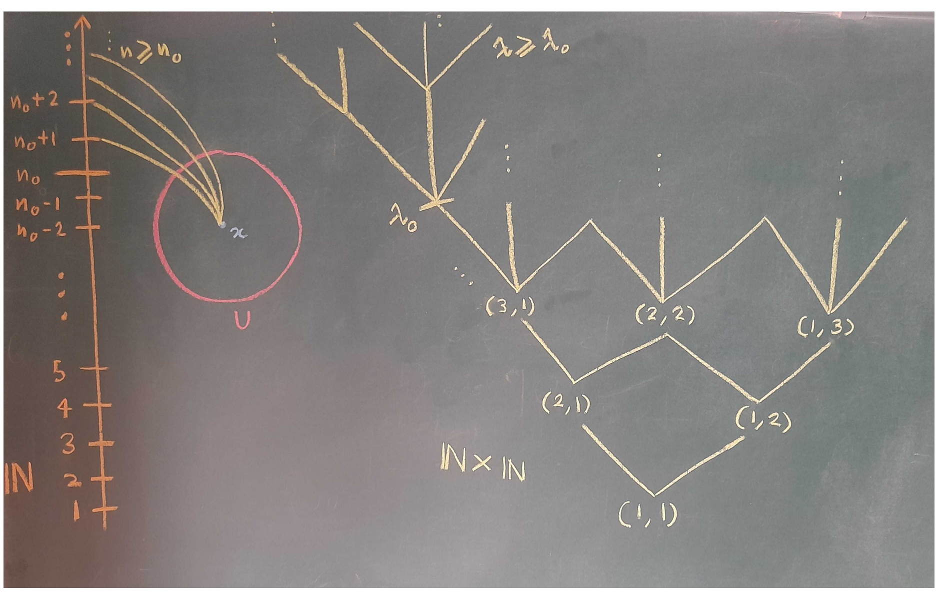

Figure 1.1: Visualisation of sequences vs nets.

Every sequence is a net, but not every net is a sequence. We want to show that the familiar results about the convergence of sequences in metric spaces also hold for nets in topological spaces.

Definition 1.9 (Net, convergence of). A net $\{s_\lambda\}_\lambda$ in X **converges** to a point $x \in X$ (written $s_\lambda \to x$) if for every open set U containing x, there exists an index $\lambda_0 \in \Lambda$ such that $s_\lambda \in U$ for all $\lambda \geqslant \lambda_0$. We then say that the net is *eventually in U*.

Proposition. Let X be a topological space and $x \in X$, then there always exists a net $\{s_\lambda\}_\lambda$ in X such that

$$s_\lambda \to x.$$

Proof. Consider a local basis $\mathcal{N}_x$ of x in X. For each[1] $U \in \mathcal{N}_x$, choose $s_U \in U$, then

$$\{s_U : U \in \mathcal{N}_x\}$$

is a net in X. Let $W \in \mathcal{T}$ s.t. $x \in W$, then in particular W itself is a nbhood of x so $W \in \mathcal{N}_x$.

Now, if $V \geqslant W$, then $s_V \in V \subset W \implies s_U \to x$. ∎

Proposition. A point $x_0 \in X$ is a *limit point* (or *cluster point*) of a subset $A \subset X$ iff for any open set U containing x_0, there exists a net $\{s_\lambda\}_\lambda$ in A (or a net $\{s_\lambda\}_\lambda$ in $A \setminus \{x_0\}$) s.t. $s_\lambda \to x_0$.

Proof. $\underline{\implies}$: $x_0 \in X$ is a limit point (cluster point) of $A \subset X$ so

$$V \in \mathcal{N}_{x_0} \implies A \cap V \neq \varnothing \ \ (A \setminus \{x_0\} \cap V \neq \varnothing).$$

Choose $s_V \in A \cap V$ $(A \setminus \{x_0\} \cap V)$. Then $\{s_V : V \in \mathcal{N}_{x_0}\}$ is a net in A $(A \setminus \{x_0\})$ s.t. $s_V \to x_0$.

$\underline{\impliedby}$: Let $\{s_\lambda : \lambda \in \Lambda\}$ be a net in A $(A \setminus \{x_0\})$ s.t. $s_\lambda \to x_0$. Choose $U \in \mathcal{T}$ s.t. $x_0 \in U$, then there exists an index $\lambda_0 \in \Lambda$ s.t. $s_\lambda \in U$ for all $\lambda \geqslant \lambda_0$. In particular, $s_{\lambda_0} \in A \cap U$ $(A \setminus \{x_0\} \cap U)$, so $s_{\lambda_0} = x_0$ is a limit point (cluster point) of A. ∎

Proposition. $f : (X, \mathcal{T}) \to (Y, \mathcal{T}')$ is *continuous* at $x_0 \in X$ iff for every net $\{s_\lambda\}_{\lambda \in \Lambda}$ in X,

$$s_\lambda \to x_0 \implies f(s_\lambda) \to f(x_0).$$

Proof. $\underline{\implies}$: Choose a $W \in \mathcal{T}'$ s.t. $f(x_0) \in W$, then by continuity of f at x_0, there exists a $V \in \mathcal{T}$ s.t. $x_0 \in V$ and $f(V) \subset W$. Now, since $s_\lambda \to x_0$, there exists an index $\lambda_0 \in \Lambda$ s.t. $s_\lambda \in V$ for all $\lambda \geqslant \lambda_0$. Then

$$f(s_\lambda) \in f(V) \subset W \quad \forall \lambda \geqslant \lambda_0 \implies f(s_\lambda) \to f(x_0).$$

$\underline{\impliedby}$: If possible let f be not continuous at $x_0 \in X$, then there exists a $W \in \mathcal{T}'$ s.t. $f(x_0) \in W$ but for all $V \in \mathcal{T}$ s.t. $x_0 \in V$, we have $f(V) \not\subset W$. In particular, for each $V \in \mathcal{N}_{x_0}$, $f(V) \not\subset W \implies f(V) \setminus W \neq \varnothing$. Then there exists an $s_V \in V$ s.t. $f(s_V) \notin W$. Then $\{s_V : V \in \mathcal{N}_{x_0}\}$ is a net in X s.t. $s_V \to x_0$, but $f(s_V) \notin W$, so $f(s_V) \not\to f(x_0)$, which is a contradiction. ∎

Theorem 1.10. A topological space $(X, \mathcal{T})$ is T_2 (*Hausdorff*) iff every convergent net in X has a unique limit.

[1] These nbhoods U will be the *indices* for our net.

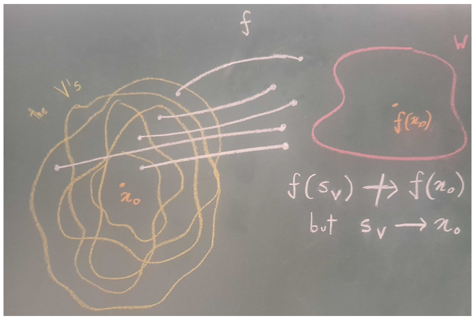

Figure 1.2: $f(s_V) \not\to f(x_0)$ although $s_V \to x_0$.

Proof. $\underline{\implies}$: If X is Hausdorff, suppose $\{s_\lambda\}_{\lambda\in\Lambda}$ is a net converging to x, y with $x \neq y$, if possible. Then there exist $U, V \in \mathcal{T}$ s.t. $x \in U$, $y \in V$ and $U \cap V = \varnothing$. Since $s_\lambda \to x$ and $s_\lambda \to y$,

$$\exists \lambda_1 \in \Lambda : s_\lambda \in U \quad \forall \lambda \geqslant \lambda_1, \quad \exists \lambda_2 \in \Lambda : s_\lambda \in V \quad \forall \lambda \geqslant \lambda_2.$$

By definition of directed set, there exists a $\lambda_0 \in \Lambda$ s.t. $\lambda_0 \geqslant \lambda_1, \lambda_2$, then $s_{\lambda_0} \in U \cap V$, which is a contradiction.

$\underline{\impliedby}$: If possible let X be not Hausdorff. Then there exists $x, y \in X$ s.t. $x \neq y$ and $U \cap V \neq \varnothing$ for all $U \in \mathcal{N}_x$, $V \in \mathcal{N}_y$. Recall that the Cartesian product of two directed sets is a directed set 1.7. Let $\Lambda = \mathcal{N}_x \times \mathcal{N}_y$ with

$$(U_1, V_1) \leqslant (U_2, V_2) \quad \text{if} \quad U_1 \supset U_2,\ V_1 \supset V_2.$$

We choose $s_{(U,V)} \in U \cap V$, then $s_{(U,V)} \to x, y$ but $x \neq y$. Contradiction. ∎

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1.2.1 Subnets

Definition 1.11 (Cluster point). $x_0 \in X$ is a **cluster point** of a net $\{s_\lambda\}_{\lambda\in\Lambda}$ if for every open set U, $x_0 \in U$ and $\lambda_0 \in \Lambda$, there exists a $\lambda \geqslant \lambda_0$ s.t. $s_\lambda \in U$. We say that the net is *cofinally* or *frequently in* U.

$s_\lambda \to x_0 \implies x_0$ is a cluster point of $\{s_\lambda\}_{\lambda\in\Lambda}$ but the converse is not true; consider the net (in fact sequence) $\{1, 2, 1, 2, 1, 2, \dots\}$ then 1 and 2 are both cluster points but the net is not convergent.

Definition 1.12 (Ultranet). $\{s_\lambda\}_{\lambda\in\Lambda}$ is an **ultranet** or **maximal net** if for any $A \subset X$ it is eventually in A or $X \setminus A$.

Theorem 1.13. If the net $\{s_\lambda\}_\lambda$ is maximal and x_0 is a cluster point of the net, then $s_\lambda \to x_0$.

Proof. Let U be an open set containing x_0. Since $\{s_\lambda\}_\lambda$ is maximal, it is eventually in U or $X \setminus U$. If it is eventually in $X \setminus U$, then there exists a λ_0 s.t. $s_\lambda \in X \setminus U$ for all $\lambda \geqslant \lambda_0$. As x_0 is a cluster point, there exists a $\lambda_1 \geqslant \lambda_0$ s.t. $s_{\lambda_1} \in U$, which means $s_{\lambda_1} \in U$ and $X \setminus U$, i.e. $s_{\lambda_1} \in U \cap (X \setminus U) = \varnothing$, which is a contradiction. Therefore, the net must be eventually in U, which automatically yields $s_\lambda \to x_0$. ∎

We want to generalise the notion of a subsequence to subnet, but any two elements of a directed set may not be comparable. So we have to generalise the notion in such a way that it does not require totally orderedness of $\mathbb{N}$.

The natural map between any two arbitrary sets A, B where $A \subset B$ is the inclusion map $\iota : A \to B$. Consider the sequence $\{x_1, x_2, x_3, x_4, x_5 \dots\}$ and a corresponding subsequence

$$\{x_1, x_4, x_5, x_9, x_{16}, \dots\} = \{y_1, y_4, y_5, y_9, y_{16}, \dots\}.$$

Let $M = \{1, 4, 5, 9, 16, \dots\} \subset \{1, 2, 3, 4, 5, \dots\} = \mathbb{N}$, then the subsequence can be expressed in terms of the original sequence indexed by the inclusion map,

$$\boxed{y_\alpha = x_{\iota(\alpha)}}, \quad \alpha \in M$$

where $x : \mathbb{N} \to X$ and $y : M \to X$. Here $\iota : M \to \mathbb{N}$ is the inclusion map. Now, given any positive integer $m \in \mathbb{N}$ the inclusion of elements from M in $\mathbb{N}$ eventually exceeds m; for example, given $m = 15 \in \mathbb{N}$, $\exists \alpha_0 = 16 \in M$ s.t. $\iota(\alpha) \geqslant m$ for all $\alpha \geqslant \alpha_0$. This holds due to the *archimedean property* of $\mathbb{N}$. Notice, however, that we haven't used totally orderedness of $\mathbb{N}$ anywhere in this characterisation.

Definition 1.14 (Subnet). Let $s : \Lambda \to X$, $t : \mathcal{M} \to X$ be nets. $\{t_\mu\}_{\mu \in \mathcal{M}}$ is a **subnet** of $\{s_\lambda\}_{\lambda \in \Lambda}$ if there exists a monotone increasing map $\iota : \mathcal{M} \to \Lambda$ s.t.

($\mathcal{M}$1) $t_\mu = s_{\iota(\mu)} \quad \forall \mu \in \mathcal{M}$

($\mathcal{M}$2) for $\lambda_0 \in \Lambda$ there exists $\mu_0 \in \mathcal{M}$ s.t. $\iota(\mu) \geqslant \lambda_0 \quad \forall \mu \geqslant \mu_0$ ($\iota(\mathcal{M})$ is *cofinal* in Λ)

Theorem 1.15. x_0 is a cluster point of $\{s_\lambda\}_{\lambda \in \Lambda}$ iff there exists a subnet $\{t_\mu\}_{\mu \in \mathcal{M}}$ of $\{s_\lambda\}$ s.t. $t_\mu \to x_0$.

Proof. $\underline{\Longleftarrow}$: Let $x_0 \in U$ for an open set U, $\lambda_0 \in \Lambda$. Then, as $t_\mu \to x_0$, there exists $\mu_1 \in \mathcal{M}$ s.t. $t_\mu \in U$ for all $\mu \geqslant \mu_1$. Moreover, from the definition of a subnet, there exists $\mu_2 \in \mathcal{M}$ s.t. $\iota(\mu) \geqslant \lambda_0 \quad \forall \mu \geqslant \mu_2$. Now, $t_\mu = s_{\iota(\mu)}$. There exists $\mu_3 \in \mathcal{M}$ s.t. $\mu_3 \geqslant \mu_1, \mu_2$ as $\mathcal{M}$ is directed. Then, $\iota(\mu_3) = \lambda \geqslant \lambda_0$ and $s_\lambda = s_{\iota(\mu_3)} \in U$. So x_0 is a cluster point of $\{s_\lambda\}$.

$\underline{\Longrightarrow}$: Let $\mathcal{M} = \mathcal{N}_{x_0} \times \Lambda = \{(U, \lambda) : U \in \mathcal{N}_{x_0}, \lambda \in \Lambda\}$, with $(U_1, \lambda_1) \geqslant (U_2, \lambda_2)$ if $U_1 \subset U_2$ and $\lambda_1 \geqslant \lambda_2$. Let $(U, \lambda) \in \mathcal{M}$. Then there exists $p = p_{(U,\lambda)} \in \Lambda$ s.t. $p \geqslant \lambda$ and $s_p \in U$. Define the map $\iota : \mathcal{M} \to \Lambda$ given by $\iota(U, \lambda) = p_{(U,\lambda)}$. Then,

$$t_{(U,\lambda)} = s_{\iota(U,\lambda)} = s_{p_{(U,\lambda)}}.$$

Let $\lambda_0 \in \Lambda$. Choose any $\mu_0 = (V, \lambda_0) \in \mathcal{M}$. Then,

$$\begin{aligned} \mu = (U, \lambda) &\geqslant (V, \lambda_0) = \mu_0 \\ \implies \iota(\mu) = p_{(U,\lambda)} &= p \geqslant \lambda \geqslant \lambda_0 \\ &\implies \iota(\mu) \geqslant \lambda_0. \end{aligned}$$

So t_μ is a subnet. What's left to show is convergence. Let W be an open set containing x_0, and choose

$$\mu' = (W, \lambda_0), \;\; \lambda_0 \in \Lambda.$$

Then,

$$\begin{aligned} \mu = (U, \lambda) &\geqslant (W, \lambda_0) = \mu' \\ \implies t_\mu &= s_{\iota(\mu)} \\ &= s_{p_{(U,\lambda)}} \in U \subset W \end{aligned}$$

where the last line is due to $U \geqslant W$. Then $t_\mu \in W$ for all $\mu \geqslant \mu'$, so $t_\mu \to x_0$. ∎

1.2.2 Filters

Definition 1.16 (Filter). A **filter** $\mathcal{F}$ on X is a nonempty subset of $\mathcal{P}(X)$ such that

($\mathcal{F}$1) $\varnothing \notin \mathcal{F}$

($\mathcal{F}$2) $A, B \in \mathcal{F} \implies A \cap B \in \mathcal{F}$

($\mathcal{F}$3) $A \in \mathcal{F}, A \subset B \implies B \in \mathcal{F}$

Example 1.17. $\mathcal{N}_{x_0}$ is a filter in any topological space.

Example 1.18. Let $x_0 \in X$ where X is a general nonempty set, not necessarily a topological space. Then

$$\mathcal{F}_{x_0} = \{A \subset X : x_0 \in A\}$$

is a filter, called the **principal filter at x_0**.

Example 1.19. The filter $\mathcal{F}_{\mathbb{N}}$ is the collection of all cofinite subsets of $\mathbb{N}$.

Example 1.20. For any infinite set X, $\mathcal{F}_X$ is the collection of all cofinite subsets of X, called the **Fréchet filter**.

Definition/Proposition 1.21 (Filter base). A subset $\mathcal{B} \subset \mathcal{F}$ is a **basis of the filter $\mathcal{F}$** (or a **filter base**) if for any $F \in \mathcal{F}$ there exists $B \in \mathcal{B}$ s.t. $B \subset F$.

$\mathcal{B} \subset \mathcal{P}(X)$ forms a filter base if

($\mathcal{FB}$1) $\varnothing \notin \mathcal{B}$

($\mathcal{FB}$2) for $B_1, B_2 \in \mathcal{B}$ there exists $B_3 \in \mathcal{B}$ s.t. $B_3 \subset B_1 \cap B_2$

We can thus write

$$\mathcal{F} = \{F \subset X : F \supset B \text{ for some } B \in \mathcal{B}\}.$$

For $\mathcal{U}_1, \mathcal{U}_2 \subset \mathcal{P}(X)$ we define $\mathcal{U}_1 \leqslant \mathcal{U}_2$ if $\mathcal{U}_1 \subset \mathcal{U}_2$. Then $\leqslant$ is a partial order relation on $\mathcal{P}(\mathcal{P}(X))$.

Definition/Proposition 1.22 (Ultrafilter). The collection of all filters on X with $\leqslant$ is a poset. A filter $\mathcal{F}^*$ is called an **ultrafilter** if it is not properly contained in any filter.

Example 1.23. If $\mathcal{F}_{x_0} \subset \mathcal{F}'$ then for any $F \in \mathcal{F}'$,

$$F, \{x_0\} \in \mathcal{F}' \implies F \cap \{x_0\} \in \mathcal{F}'.$$

So $F \cap \{x_0\} \neq \varnothing \implies x_0 \in F \implies F \in \mathcal{F}_{x_0}$. Hence, $\mathcal{F}_{x_0} = \mathcal{F}'$ so the principal filter at x_0 is an ultrafilter.

Ultrafilters are analogous to components from connectedness. Indeed, for any topological space the connected components are the *maximal connected subsets* just as ultrafilters are the maximal proper filters.

Bibliography

[Tol69] Leo Tolstoy. *War and Peace*. Penguin Classics, 1869.

List of Definitions